

JavaScript

- Objects
- JSON
- LocalStorage

What is an Object?

- A collection of key-value pairs
- Example:

```
const student = {  
  name: "Ali",  
  mark: 90  
};
```

- name → key
- "Ali" → value

Why Use Objects?

- Objects help us store **related data together**
- Without object:

```
let name = "Ali";  
let mark = 90;
```

Why Use Objects?

- Why Use Objects?

```
const student = {  
  name: "Ali",  
  mark: 90  
};
```

- Cleaner
- Organized
- Scalable

Access Object Data

```
const student = {  
  name: "Ali",  
  mark: 90  
};
```

- Access values:

```
console.log(student.name);    // Ali  
console.log(student["mark"]); // 90
```

Modify Object

```
student.mark = 95;  
student.age = 20;  
  
console.log(student);
```

- Result

```
{  
  name: "Ali",  
  mark: 95,  
  age: 20  
}
```

Objects in Arrays (Real Use)

```
const students = [  
  { name: "Ali", mark: 90 },  
  { name: "Sara", mark: 85 }  
];
```

- Access

```
console.log(students[0].name); // Ali
```

What is JSON?

- **JSON = JavaScript Object Notation**
- **Used to store and exchange data**
- **Looks like an object, but it is a string format**

Object vs JSON

- Object

```
const student = {  
  name: "Ali",  
  mark: 90  
};
```

- JSON

```
{  
  "name": "Ali",  
  "mark": 90  
}
```

- Must use **double quotes**
- Keys must be strings

JSON.stringify()

- JSON.stringify()

```
const student = { name: "Ali", mark: 90 };  
  
const json = JSON.stringify(student);  
  
console.log(json);
```

- Output

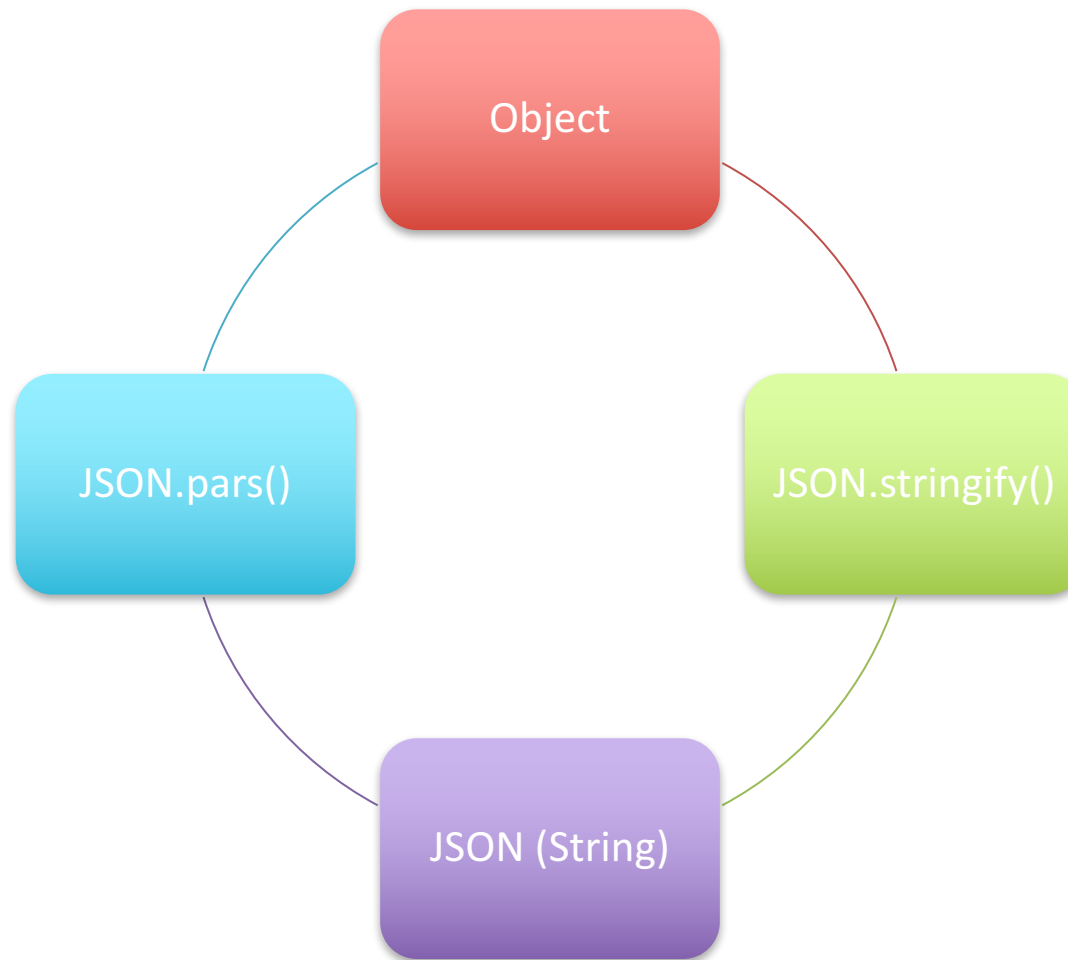
```
{"name":"Ali","mark":90}
```

JSON.parse()

- Convert JSON → Object

```
const text = '{"name":"Ali","mark":90}';  
  
const obj = JSON.parse(text);  
  
console.log(obj.name); // Ali
```

JSON Flow



What is LocalStorage?

- LocalStorage is a browser storage to **save data permanently**
 - Data stays after refresh
 - Data stored as **string**

LocalStorage Basic Methods

```
localStorage.setItem("name", "Ali"); // save
```

```
localStorage.getItem("name"); // get
```

```
localStorage.removeItem("name"); // delete
```

```
localStorage.clear(); // delete all
```

Example (Simple)

```
localStorage.setItem("city", "Berlin");  
console.log(localStorage.getItem("city"));
```

- Output

Berlin

Store Objects in LocalStorage

- LocalStorage only stores strings

```
const student = {  
  name: "Sara",  
  mark: 95  
};  
  
localStorage.setItem("student", JSON.stringify(student));
```

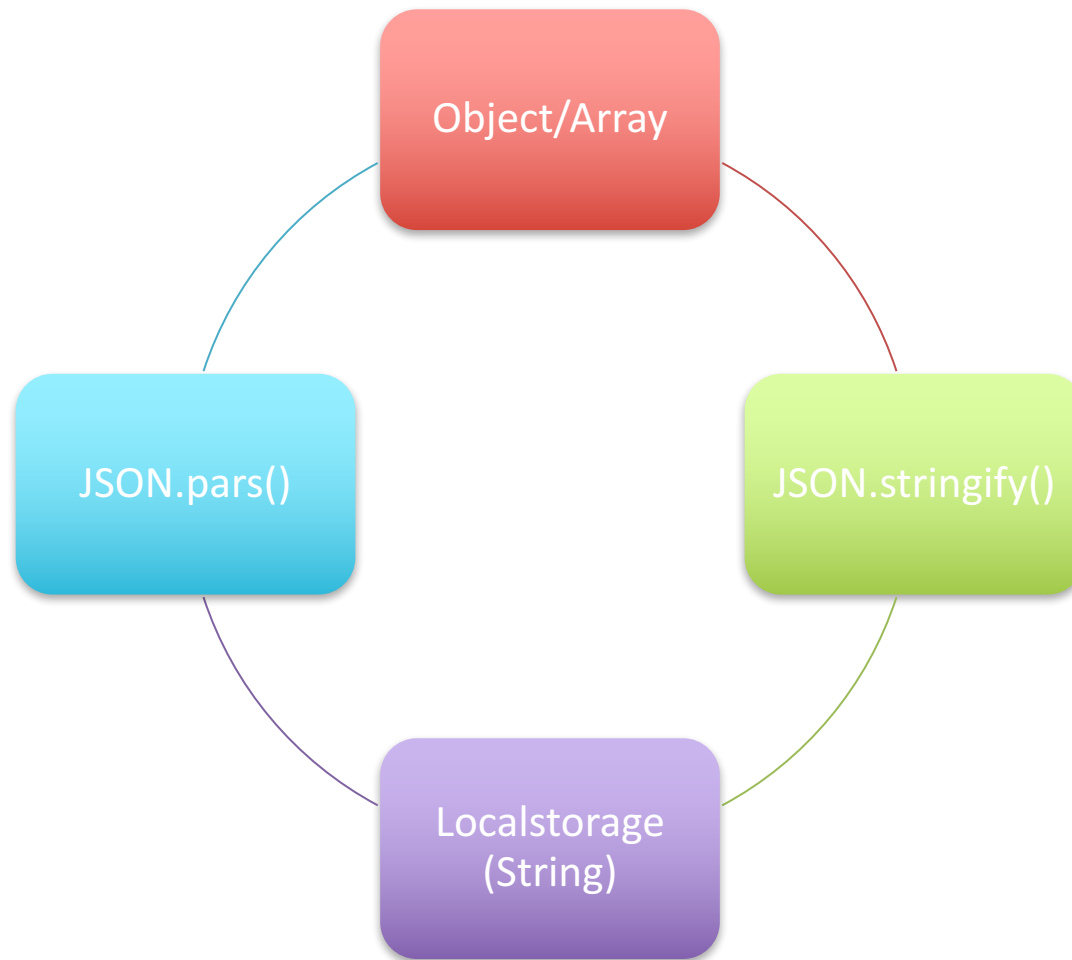

Retrieve Object

```
const data = JSON.parse(localStorage.getItem("student"));  
  
console.log(data.name); // Sara
```

Real Example (Tasks)

```
const tasks = [  
  { name: "Study JS", completed: false }  
];  
  
// Save  
localStorage.setItem("tasks", JSON.stringify(tasks));  
  
// Load  
const savedTasks = JSON.parse(localStorage.getItem("tasks"));
```

Full Flow



Common Mistake

- Wrong

```
localStorage.setItem("tasks", tasks);
```

- Correct

```
localStorage.setItem("tasks", JSON.stringify(tasks));
```

Summary

- **Objects**
 - Store structured data
 - Key-value pairs
- **JSON**
 - Format for data exchange
 - Use stringify() and parse()
- **LocalStorage**
 - Saves data in browser
 - Works with JSON